Hyperliquid Trader Behavior Analysis Using Bitcoin Market Sentiment

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1. Objective

The objective of this project was to analyze the relationship between Bitcoin market sentiment (Fear & Greed Index) and trader behavior on the Hyperliquid exchange. By combining daily market sentiment data with historical trading records, the project aimed to identify patterns in trading performance, behavioral changes across different market conditions, and actionable strategies that could support more informed trading decisions.

2. Methodology

The analysis was carried out through the following steps:

- Cleaned and validated both datasets by checking missing values, duplicates, and data types.
- Converted timestamps into daily dates and merged trader data with Bitcoin market sentiment.
- Engineered key metrics including: Daily realized PnL, Win Rate, Average Trade Size, Trading Frequency, Long/Short Ratio, Drawdown Proxy
- Compared trader performance across different sentiment categories.
- Segmented traders based on trading frequency, profitability, and win rate.
- Applied K-Means Clustering to identify distinct trader archetypes using behavioral features.

3. Key Findings

- **Extreme Greed** recorded the highest average realized PnL and win rate, while **Extreme Fear** exhibited the lowest win rate and the highest average losses.
- **Fear** periods showed the highest trading activity, the largest average trade sizes, and a strong long-position bias, indicating increased market participation during uncertain conditions.
- Frequent traders generated more than three times the cumulative realized PnL of infrequent traders, suggesting that active participation was associated with stronger overall profitability.
- A higher win rate did not necessarily correspond to greater profits, emphasizing the importance of trade quality and effective risk management over simply maximizing winning trades.
- K-Means clustering identified four distinct trader archetypes, demonstrating that successful traders followed different strategies, including high-frequency trading, larger position sizing, and disciplined execution.

4. Strategy Recommendations

- During **Extreme Fear**, traders should prioritize capital preservation by reducing position sizes and applying stricter risk management practices.
- During **Extreme Greed**, traders may maintain normal or moderately increased exposure while remaining cautious of potential market reversals.
- Focus on improving **risk-reward ratios** rather than solely increasing win rates, as profitability was influenced more by trade quality than by winning percentage.
- Develop a disciplined trading plan that balances participation, position sizing, and risk management instead of relying on a single performance metric.